# Hermenegildo Mashele

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## COVER LETTER

Dear Hiring Manager,

I am writing to express my interest in graduate and junior opportunities in Artificial Intelligence, Machine Learning, Data Science, and Software Engineering. I am currently completing a BSc Honours in Computer Science, specialising in Artificial Intelligence, at the University of Johannesburg. My studies and projects have given me practical experience in machine learning, deep learning, computer vision, optimisation, reinforcement learning, and software development.

My main honours research project, **Improving Long-Term Simulation Accuracy through Self-Correcting World Models**, focused on reducing prediction drift in learned simulations. I designed and implemented an Encoder-GRU-Decoder world model in Python and PyTorch and compared open-loop, fixed-interval, and adaptive correction strategies in Atari Breakout. The adaptive strategy reduced average prediction error by approximately 37% compared with the open-loop baseline while requiring fewer corrections than the fixed strategy. This project strengthened my skills in model implementation, controlled experimentation, evaluation, debugging, and critical analysis of model behaviour.

I have also completed projects in self-supervised contrastive learning, wildlife image re-identification, genetic algorithms, procedural content generation, reinforcement learning, graphics programming, and similarity search. Through these projects I have worked with technologies including Python, C++, PyTorch, ResNet architectures, HNSW, Godot, OpenGL, Vulkan, and SDL. I enjoy learning by building and am comfortable working through unfamiliar technical problems step by step.

In addition to my technical work, I serve as a tutor in the Academy of Computer Science and Software Engineering at the University of Johannesburg. This role has strengthened my communication, problem-solving, and teamwork skills by requiring me to explain technical concepts clearly and support students with programming and practical coursework.

I am looking for an opportunity where I can contribute my technical foundation, research experience, and willingness to learn while continuing to grow as a software and AI professional. I would value the opportunity to discuss how my background and skills could contribute to your organisation.

Thank you for considering my application. I look forward to the opportunity to contribute and grow within your team.

Kind regards,
**Hermenegildo Mashele**